

The norm induced by the standard inner product on $\mathbb{R}^2$ is the ℓ_2 norm.

■ Question 28.

□

What is $\|\vec{v}\|$ if the inner product is defined as [question 27](#)?

§2.2 Best approximation

Let $(\mathbb{X}, d)$ be a metric space and let $\mathcal{U}$ be a subspace of $\mathbb{X}$. An approximation method is a mapping from $\mathbb{X}$ to $\mathcal{U}$. In other words, given any $f \in \mathbb{X}$, the method picks the element $u_f \in \mathcal{U}$, which is regarded as an approximation to f . To find whether the method is good or bad, it should be compared with the best approximation.

Definition 2.2.13

A best approximation to $f \in \mathbb{X}$ from $\mathcal{U}$ is an element $u^* \in \mathcal{U}$ s.t.

$$d(f, u^*) = \inf\{d(f, u) : u \in \mathcal{U}\} =: \text{dist}(f, \mathcal{U})$$

Regarding the element of best approximation, basic questions are (i) existence, (ii) uniqueness, (iii) characterization: how one can recognize u^* other than comparing it with all other elements of $\mathcal{U}$, this is important for (iv) construction.

In our case, $\mathbb{X} = C^0(I)$ and $\mathcal{U} = \mathcal{P}_n$. First, let's consider the ℓ_∞ norm on $C^0(I)$ so that $d(f, g) = \|f - g\|_\infty$.

Definition 2.2.14

Let $f \in C^0(I)$. Then $p_* \in \mathcal{P}_n$ is a best approximation to f provided that

$$\|f - p_*\|_{\infty, I} = \inf\{\|f - p\|_{\infty, I} \mid p \in \mathcal{P}_n\}$$

■ Question 29.

□

Prove that p_* exists.

Note: While we will not prove it here, it turns out that p_* is unique in this norm. This is somewhat surprising, as uniqueness is not always guaranteed for general subspaces of $C^0(I)$.

■ Question 30.

□

Show that the best approximation (using the ℓ_∞ norm) of $f \in C^0(I)$ by a constant c is

$$c = \frac{1}{2} \left(\inf_{x \in I} f(x) + \sup_{x \in I} f(x) \right).$$

Another way to define “best” approximation is using a norm induced from an inner product structure on $C[a, b]$. This will be analogous to ℓ_2 norm in the discrete case.

2.2.1 Another detour - Orthogonal Polynomials

Definition 2.2.15: Inner Product

We can define an inner product of the functions $f, g \in C[a, b]$ as

$$\langle f, g \rangle = \int_a^b f(x)g(x)w(x)dx.$$

for any choice of weight function $w(x)$ that is positive, continuous, and integrable on (a, b) .

Note: The purpose of the weight function will be to assign varying degrees of importance to errors on different portions of the interval. This will be relevant later when talking about quadrature.

With this definition in mind, we say $f, g \in C[a, b]$ are orthogonal w.r.t. a given inner product if and only if $\langle f, g \rangle = 0$.

■ Question 31.

Prove that the Chebyshev polynomials $\omega_n = \frac{T_n(x)}{2^{n-1}}$ are orthogonal on $[-1, 1]$ with the weight $w(x) = \frac{1}{\sqrt{1-x^2}}$. □

The following theorem is a consequence of the **Gram-Schmidt algorithm** you have seen before.

Theorem 2.2.16

Given a finite-dimensional inner product space with a set of linearly independent vectors $\{\beta_1, \dots, \beta_n\}$, we can always construct a set of orthogonal vectors $\{\alpha_1, \dots, \alpha_n\}$ so that $\{\alpha_1, \dots, \alpha_i\}$ is a basis for the $\text{span}\{\beta_1, \dots, \beta_i\}$ for $i = 1, 2, \dots, n$.

In particular, this means we can apply Gram-Schmidt to the monomial basis to construct a family of orthogonal polynomials $\{\varphi_n\}_{n \geq 0}$ that satisfy:

- $\varphi_n \in \mathcal{P}_n$ and $\deg(\varphi_n) = n$.
- $\{\varphi_0, \varphi_1, \dots, \varphi_i\}$ forms a basis for $\mathcal{P}_n$
- $\langle \varphi_n, p \rangle = 0$ for $p \in \mathcal{P}_{n-1}$

Furthermore, if we specify that each φ_n is monic, then the family is unique.

Theorem 2.2.17: Three Term Recurrence for Orthogonal Polynomials

Given a weight function $w(x)$, a real interval $[a, b]$, and an associated inner product

$$\langle f, g \rangle = \int_a^b f(x)g(x)w(x)dx$$

a family of (monic) orthogonal polynomials $\{\varphi_k\}_{k=0}^n$ can be generated as follows:

$$\begin{aligned}\varphi_0(x) &\equiv 1 \\ \varphi_1(x) &= (x - \alpha_0) \varphi_0(x), \\ \varphi_{n+1}(x) &= (x - \alpha_n) \varphi_n(x) - \beta_n \varphi_{n-1}(x), \quad n = 1, 2, \dots,\end{aligned}$$

where

$$\alpha_n = \frac{\langle \varphi_n, x \varphi_n \rangle}{\langle \varphi_n, \varphi_n \rangle}, \quad \beta_n = \frac{\langle \varphi_n, \varphi_n \rangle}{\langle \varphi_{n-1}, \varphi_{n-1} \rangle} > 0$$

■ Question 32.

□

Check that the Chebyshev polynomials $T_n(x)$ satisfy the required recurrence relation,

$$T_{n+1}(x) = 2xT_n(x) - T_{n-1}(x)$$

and as such, $\omega_n(x)$ are the unique monic orthogonal family with weight $1/\sqrt{1-x^2}$ on $[-1, 1]$.

Example 2.2.18: Legendre Polynomials

There are several different important families of orthogonal polynomials which are useful in Numerical Analysis. One example is the family of *Legendre polynomials* named after the French mathematician Adrien-Marie Legendre. These are generated by the inner product with $w(x) \equiv 1$ on the interval $(-1, 1)$.

After some simplifications, the recurrence relations become

$$\varphi_k(x) = x\varphi_{k-1}(x) - \frac{(k-1)^2}{4(k-1)^2 - 1} \varphi_{k-2}(x) \quad \text{for } k \geq 2.$$

The first few Legendre polynomials are

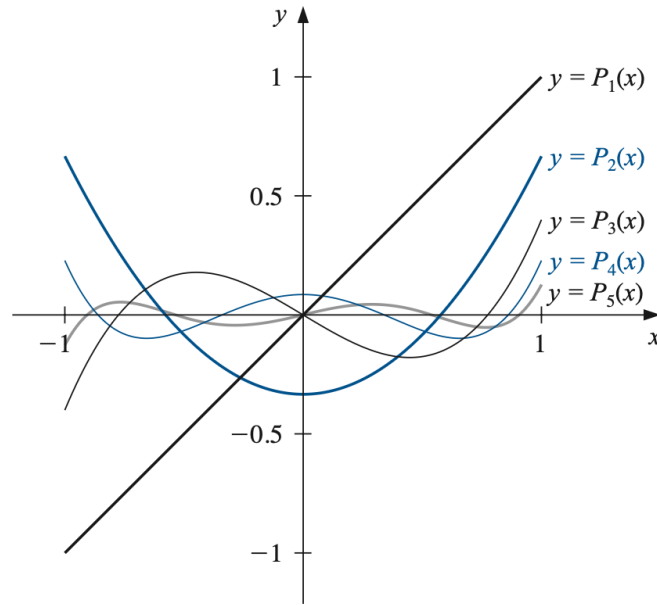
$$\begin{aligned}\varphi_0(x) &= 1 \\ \varphi_1(x) &= x \\ \varphi_2(x) &= x^2 - \frac{1}{3} \\ \varphi_3(x) &= x^3 - \frac{3}{5}x \\ \varphi_4(x) &= x^4 - \frac{6}{7}x^2 + \frac{3}{35} \\ \varphi_5(x) &= x^5 - \frac{10}{9}x^3 + \frac{5}{21}x \\ \varphi_6(x) &= x^6 - \frac{15}{11}x^4 + \frac{5}{11}x^2 - \frac{5}{231}.\end{aligned}$$

You can find graphs of the Legendre polynomials (up to $n = 5$) below in [figure 2.1](#).

■ Question 33.

□

Show that for any $w(x)$, the degree n polynomial φ_n has n **distinct** real roots.

Figure 2.1: Graphs of Legendre polynomials (up to $n = 5$)

Sketch of Proof. Suppose not. Let q be a polynomial that has the same roots as $\varphi(n)$ but with multiplicity 1. What is $\langle \varphi_n(x), q(x) \rangle$?

Conclusion

With this procedure, we can easily increase the degree of the approximating polynomial. To increase the degree by one, simply add

$$\langle f, \varphi_{n+1} \rangle \varphi_{n+1}$$

Instead, if we were to use a non-orthogonal basis (e.g. the monomial basis), to increase the degree of the approximation we would need to solve a new $(n+2)$ -by- $(n+2)$ linear system to find the coefficients of the least squares approximation.

2.2.2 Back to Least square approximation

Let's return to the original question of finding the "best approximation". With the inner product definition in mind, we define the ' L^2 norm' on $C[a, b]$ as follows:

$$\|f\|_2 := \sqrt{\langle f, f \rangle} = \left(\int_a^b f(x)^2 w(x) dx \right)^{1/2}.$$

The ' L ' stands for Lebesgue, coming from the fact that this inner product can be generalized from $C[a, b]$ to the set of all functions that are square-integrable, in the sense of Lebesgue integration. By restricting our attention to continuous functions, we dodge the measure-theoretic complexities.

The continuous least squares problem is to find the polynomial $p^* \in \mathcal{P}_n$ that minimizes $\|p - f\|_2$ among all $p \in \mathcal{P}_n$.

■ **Question 34.**

□

Given $f \in C[a, b]$, the polynomial $p^* \in \mathcal{P}_n$ minimizes $\|p - f\|_2$ among all $p \in \mathcal{P}_n$ if and only if

$$\langle p^* - f, p \rangle = 0 \quad \text{for all } p \in \mathcal{P}_n$$

Note: In other words, p^* is the b.a. iff $p^* - f \in \mathcal{P}_n^\perp$.

Theorem 2.2.19

Let $\{\varphi_k\}_{k=0}^n$ be monic polynomials orthogonal with respect to a given inner product, where $p_k \in \mathcal{P}_k$. Then the least squares approximant to any $f \in C[a, b]$ from $\mathcal{P}_n$ is given by the projection

$$L_n f = \sum_{k=0}^n \langle f, \varphi_k \rangle \varphi_k$$

In other words,

$$\|f - L_n f\|_2 = \min_{p \in \mathcal{P}_n} \|f - p\|_2$$

The proof is pretty straightforward. We just need to check that $\langle L_n f - f, \varphi_k \rangle = 0$ for all k .

■ **Question 35.**

□

Find the $p_1(x) = c_0 + c_1 x$ that best approximates $f(x) = \sin(\pi x)$ on the interval $[0, 1]$ with weight function $w(x) = 1$.